

Sample OS – Midterm Exam

For Review ONLY

Good Luck!

I. (30 points, 3 points each) Multiple Choice Questions – For each of the following questions, please select the best answer from the given choices. Remember there is only **ONE** best answer to each question.

1. Regarding multi-programming, which of the following statement is Right?
 - a) An executable program that is composed of modules built using different programming languages.
 - b) Having multiple processors which execute different programs at the same time.
 - c) Keeping several programs in memory at once and switching between them.
 - d) Each program is blocked for I/O operations and executed alternately.
2. When an interrupt is enabled, then where does the pointer moves immediately after this interrupt has occurred?
 - a) to the next instruction which is to be executed.
 - b) to the first instruction of Interrupt Service Routine.
 - c) to a fixed location in memory called interrupt vector table.
 - d) to the end of the program.
3. Which of the following statement is False about deadlock detection policies?
 - I. Wait-die scheme is a non-preemptive technique while wound-wait scheme is a preemptive technique for deadlock prevention.
 - II. If T_i requests a lock held by T_j and T_i is older than T_j then in wait-die scheme T_i will wait and in wound-wait scheme T_j will wait.
 - III. There is deadlock in the system if and only if there exists a cycle in the wait-for-graph.
 - a) I
 - b) II
 - c) I and II
 - d) II and III
4. If all processes are I/O bounded, the ready queue will almost always be _____ and the short-term scheduler will have a _____ to do:
 - a) full, little.
 - b) full, lot.

- c) empty, little.
 - d) empty, lot.
5. Which of the following statement is False?
- a) Progress does not imply Bounded Waiting.
 - b) Bounded Waiting does not imply Progress.
 - c) Starvation-freedom implies deadlock-freedom.
 - d) Starvation-freedom implies bounded-waiting.
6. Suppose that a process is in “Blocked” state waiting for some I/O operation. When the I/O operation is completed, it goes to the:
- a) Running state
 - b) Ready state
 - c) Suspended state
 - d) Terminated state.
7. What’s the biggest problem with spinlocks?
- a) They cannot guarantee bounded waiting.
 - b) They can only be used to solve two processes’ synchronization problem.
 - c) They waste CPU resources.
 - d) They rely on kernel support and cannot be implemented at user level.
8. Which of the following statement does not apply to message passing in inter-process communication?
- a) There are no conflicts needed to be avoided.
 - b) Easier to implement in distributed system.
 - c) Typically is implemented using system calls.
 - d) Faster than shared memory method.
9. At a particular time of computation, the value of a counting semaphore is 7. Then 20 wait() operations and 15 signal() operations were completed on this semaphore. The resulting value of the semaphore is:
- a) 42
 - b) 2
 - c) 7
 - d) 12

10. ' m ' processes share ' n ' resources of the same type. The maximum need of each process doesn't exceed ' n ' and the sum of all their maximum needs is always less than $m + n$. In this setup, deadlock _____:
- a) can never occur
 - b) may occur
 - c) has to occur
 - d) none of the mentioned

II. (70 points) – Write all of your answers directly on this paper.

1. (5 points) A process executes the flowing code

fork ();

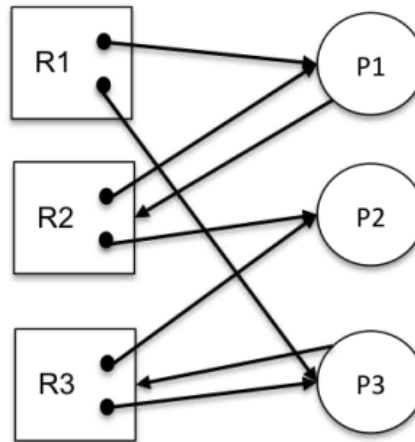
fork ();

fork ();

fork ();

What is the total number of child processes created?

2. (5 points) Consider the following resource allocation graph.



- a) Does the above allocation graph contain a deadlock (1 points)?
- b) Assume now that P2 also demands resource R1. Does this allocation graph contain a deadlock (2 points)?
- c) Assume the allocation graph at point b), and, in addition, assume that R2 has now three instances. Does this allocation graph contain a deadlock (2 points)?

3. (10 points) Two processes, P1 and P2, need to access critical sections. Consider the following synchronization construct used by the processes: here **wants1** and **wants2** are shared variables, which are initialized to be false. Please analyze whether mutual exclusion, progress and bounded waiting are satisfied or not.

<pre>/* P1 */ while (true) { wants1 = true; while (wants2 == true); /* Critical Section */ wants1 = false; } /* Remainder section */</pre>	<pre>/* P2 */ while (true) { wants2 = true; while (wants1 == true); /* Critical Section */ wants2 = false; } /* Remainder section */</pre>
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4. (10 points) Consider two processes P1 and P2. Let the period of P1 be $p1 = 50$ and the processing time of P1 be $t1 = 25$; Let the period of P2 be $p2 = 75$ and the processing time of P2 be $t2 = 30$. The **deadline** of both P1 and P2 is to complete its CPU burst by the start of its next period. If the CPU scheduling algorithm is Earliest Deadline First Scheduling (EDF), please draw the Gantt chart of scheduling and calculate the average waiting time.

5. (10 points) Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned. Please analyze whether mutual exclusion, progress and bounded waiting are satisfied or not.

Method used by P1:

while (S1 == S2);

Critical Section

S1 = S2;

Method used by P2:

while (S1 != S2);

Critical Section

S2 = !S1;

6. (10 points) The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as $S_0 = 1$, $S_1 = 0$ and $S_2 = 0$. How many times will process P0 print '0'?

Process P0

```
while (true)
{
    wait (S0);
    print '0';
    release (S1);
    print '0';
    release (S2);
}
```

Process P1

```
wait (S1);
release (S0);
```

Process P2

```
wait (S2);
release (S0);
```

7. (10 points) Consider a system that contains five processes P₀, P₁, P₂, P₃, P₄ and the three resource types A, B and C. Given the following snapshot of a system, based on banker's algorithm, is the system in a safe state?

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	1	1	2	4	3	3	2	1	0
P ₁	2	1	2	3	2	2			
P ₂	4	0	1	9	0	2			
P ₃	0	2	0	7	5	3			
P ₄	1	1	2	1	1	2			

8. (10 points) Consider the two functions **incr** and **decr** shown below.

incr(){	decr(){
wait(s);	wait(s);
X = X+1;	X = X-1;
signal(s);	signal(s);
}	}

There are 5 processes each invoking incr once, and 3 processes each invoking decr once, on the same shared variable X. The initial value of X is 10. Suppose there are two implementations of the semaphore s, as follows:

I-1: s is a binary semaphore initialized to 1.

I-2: s is a counting semaphore initialized to 2.

Let V1, V2 be the values of X at the end of execution of all the threads with implementations I-1, I-2, respectively. What is the minimum possible value of V1, V2, respectively?